

Options on a Cointegrated Spread: Pricing and Regime-Aware Hedging of a Pairs Trade

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Overview

The classic pairs trade holds long/short shares of two cointegrated stocks and bets that their spread mean-reverts. This project expresses and manages the same trade with **options on the spread**, which raises three quantitative problems. First, a European spread option with payoff $\max(S_1(T) - hS_2(T) - K, 0)$ has no Black–Scholes closed form for $K \neq 0$, because the difference of two lognormals is not lognormal; it must be priced by Monte Carlo on correlated GBM paths. Second, the $K = 0$ case *does* have a closed form which come from the Margrabe exchange-option formula, and is highly correlated with the $K \neq 0$ option, making it a natural control variate. Third, the option carries two deltas and its correlation exposure is exactly what breaks down in market stress, motivating a 2-state Markov regime-switching model and a test of regime-aware versus constant-parameter hedging.

The project runs the full pipeline: pair selection, pricing, hedging, regime modeling, and an out-of-sample backtest.

Pair selection and signal (Notebook 01)

Roughly 55 candidate pairs were screened on training data only (2021-01 to 2024-07) under a predetermined rule requiring a structural economic link. **Altria / Philip Morris (MO/PM)**, two halves of a 2008 corporate split, was selected with Engle–Granger $p = 0.015$ (ADF cross-check $p = 0.003$), hedge ratio $\beta = 0.735$, and an Ornstein–Uhlenbeck half-life of ≈ 25 days. Across training sub-windows β ranged 0.66–0.86 and the half-life 12–29 days; moderate stability, a risk the frozen-parameter design deliberately carries. The trading signal standardizes the spread, $z = (X - \theta)/\sigma$, entering at $|z| > 2$ and exiting at $z = 0$ or after a 50-day maximum hold (twice the half-life).

Pricing with a Margrabe control variate (Notebook 02)

The traded option is struck at the money in the spread, $K = (1 - \beta)S_1(0) \approx 10.60$, with maturity $T \approx 0.2y$ matched to the holding period. Monte Carlo pricing uses the exact Margrabe $K = 0$ price as a control variate: with $\text{corr}(P, C) \approx 0.85$, variance falls by roughly **4× at the traded ATM strike** ($\approx 20\times$ deep in the money, exact at $K = 0$, fading to $\approx 1.1\times$ deep out of the money). The ATM strike sits ≈ 3.4 spread-sigmas above zero, so the control behaves like a forward, capturing delta-like variation but not convexity. The resulting premium is $V_0 \approx 1.29$ per option.

Two-asset delta hedging (Notebook 03)

Because no closed form exists at $K \neq 0$, deltas come from central finite differences on the Monte Carlo price with common random numbers. The true ATM deltas are $(+0.53, -0.50)$, while the Margrabe $K = 0$ proxy pins them at ± 1 (using the proxy would double-size every hedge). Hedging a short ATM option at frequencies {none, weekly, daily} shrinks the P&L standard deviation $1.93 \rightarrow 0.33 \rightarrow 0.15$; the weekly/daily ratio of **2.15 matches the discrete-hedging $\sqrt{\Delta t}$ law** ($\sqrt{5} \approx 2.24$). All means are ≈ 0 : hedging reshapes the distribution, not the expectation.

Regime switching: pricing risk vs hedging value (Notebook 04)

A 2-state (calm/volatile) Markov model estimated on training returns finds materially higher volatilities in the volatile regime with similar correlations (0.60 vs 0.52), so the mixture acts mainly through the vol level.

Pricing: the regime-switching vs constant-parameter price gap is -2.1% of premium at the ATM strike but **+32% deep out of the money**, where the volatile regime's fat tail dominates — model risk grows with strike.

Hedging: a hedger that detects the regime from 21-day trailing realized vol is statistically indistinguishable from a naive constant-parameter hedger (P&L std 0.478 vs 0.480). This is a negative result with a measured mechanism: the volatile regime lasts 15.3 days on average, shorter than the 21-day detection window, so detection arrives roughly one regime-lifetime late. Regime information matters for pricing and risk measurement, not for delta hedging at this horizon.

Out-of-sample backtest: shares vs options (Notebook 05)

Every parameter is frozen from training; the test window is 2024-07 to 2026-06. The signal fired four times: one clean mean-reversion win ($z: -2.04 \rightarrow 0.09$), one partial reversion at timeout ($2.08 \rightarrow 0.56$), and two adverse divergences

(2.04 $\rightarrow$ 2.20; 2.13 $\rightarrow$ 3.10, censored at data end). Consistent with test-window diagnostics (EG $p = 0.63$; spread mean $+0.74\sigma$ above the frozen θ), the cointegration weakened out of sample, and the backtest shows what each vehicle does when that happens. Through mean reversion the vehicles behave similarly, with options lagging by time-value decay (total P&L: shares 3.00 vs options 1.68 BS-marked / 1.47 RS-marked). Through divergence the difference is structural: the share book loses roughly linearly and unboundedly (worst trade -1.62 with the spread still rising), while the **option's loss is capped at its premium** (worst trade -2.15 to -2.42 against entry premiums of 1.76 – 2.15). That risk cap is the economic argument for the option vehicle. The BS-marked and RS-marked books hold identical positions; their $\approx \$0.05$ -per-trade gap is valuation model risk, not a strategy difference.

Limitations and extensions

The results are presented with explicit statistical honesty. Screening ran ~ 55 tests at the 5% level (≈ 2.9 expected false discoveries), only partially mitigated by the predetermined rule and economic-link requirement. Four trades, one censored, permit no inference, the backtest demonstrates mechanics, not expected profitability. Spread options are model-marked rather than market-marked, dividends are omitted (shifting the forward spread by $\approx 0.15\sigma$ and deltas by $\approx 2\%$), transaction costs are excluded, regime detection uses a threshold rule rather than filtered probabilities, and parameters are frozen rather than re-estimated. Natural extensions include cointegration-consistent option pricing (Duan & Pliska 2004), a convexity-matched control variate (Kirk or geometric-average spread option), filtered regime probabilities via HMM/EM (Hamilton 1989), and rolling re-estimation.

NOTE on AI use: AI assistance was used for code review, debugging, simplifying the notebook structure, and revising explanatory text. All modeling choices, empirical analysis, interpretation of results, and final project decisions were reviewed and completed by the author.